spent: calculations using this mathematical framework agree with experimental findings to an unparalleled precision (e.g., calculations of the effect of vacuum fluctuations on the magnetic properties of electrons agree with experimental results to one part in a billion).[5]

Yet despite all this success, for many decades physicists have been aware that quantum jitters have been fomenting discontent within the laws of physics.

Jitters and Their Discontent [6]

So far, we've discussed only quantum jitters for fields that exist *within* space. What about the quantum jitters of space itself? While this might sound mysterious, it's actually just another example of quantum field jitters—an example, however, that proves particularly troublesome. In the general theory of relativity, Einstein established that the gravitational force can be described by warps and curves in the fabric of space; he showed that gravitational fields manifest themselves through the shape or geometry of space (and of spacetime, more generally). Now, just like any other field, the gravitational field is subject to quantum jitters: the uncertainty principle ensures that over tiny distance scales, the gravitational field fluctuates up and down. And since the gravitational field is synonymous with the shape of space, such quantum jitters mean that the shape of space fluctuates randomly. Again, as with all examples of quantum uncertainty, on everyday distance scales the jitters are too small to be sensed directly, and the surrounding environment appears smooth, placid, and predictable. But the smaller the scale of observation the larger the uncertainty, and the more tumultuous the quantum fluctuations become.

This is illustrated in Figure 12.2, in which we sequentially magnify the fabric of space to reveal its structure at ever smaller distances. The lowermost level of the figure shows the quantum undulations of space on familiar scales and, as you can see, there's nothing to see—the undulations are unobservably small, so space appears calm and flat. But as we home in by sequentially magnifying the region, we see that the undulations of space get increasingly frenetic. By the highest level in the figure, which shows the fabric of space on scales smaller than the *Planck*